

Electron Probabilities

Part I: The Wave Equation

$$1.1 \quad \frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

$$y = A \sin(kx - \omega t)$$

$$\frac{\partial^2 y}{\partial x^2} = -k^2 A \sin(kx - \omega t)$$

$$\frac{\partial^2 y}{\partial t^2} = -\omega^2 A \sin(kx - \omega t)$$

This is a solution if $v = \frac{k}{\omega}$

In complex form:

$$y = A e^{i(kx - \omega t)}$$

$$\frac{\partial^2 y}{\partial x^2} = -k^2 A e^{i(kx - \omega t)}$$

$$\frac{\partial^2 y}{\partial t^2} = -\omega^2 A e^{i(kx - \omega t)}$$

Same solution, but differentiating exponentials is easier because you just multiply by the coefficient in the exponent, do not need to switch sin and cos.

A controls the amplitude of oscillation of the string. If it were a probability amplitude, $|A|^2$ would represent a probability.

$$1.2 \quad p = \hbar k \quad k = \frac{p}{\hbar}$$

$$E = \hbar \omega \quad \omega = \frac{E}{\hbar}$$

$$\psi(x, t) = A e^{\frac{i}{\hbar}(px - Et)}$$

$$\frac{\partial \psi}{\partial x} = \frac{p}{\hbar} \cdot \psi$$

$$\frac{\hbar}{i} \cdot \frac{\partial \psi}{\partial x} = p \cdot \psi$$

$$\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} E \psi \quad E \cdot \psi = -\frac{\hbar}{i} \frac{\partial \psi}{\partial t}$$

We want operators that return p or E as an eigenvalue when applied to ψ .

$$\begin{aligned} i &= e^{i\frac{\pi}{2}} \\ i^{-1} &= e^{-i\frac{\pi}{2}} \\ &= e^{\frac{3\pi}{2}i} \\ &= -i \end{aligned}$$

$$p \cdot \psi = -i\hbar \frac{\partial \psi}{\partial x}$$

$$E \cdot \psi = i\hbar \frac{\partial \psi}{\partial t}$$

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}$$

$$\hat{E} = i\hbar \frac{\partial}{\partial t}$$

For a free particle, $E = \frac{p^2}{2m}$. For operators:

$$\hat{E}\psi = \frac{\hat{p}^2}{2m}\psi$$

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-i\hbar \frac{\partial \psi}{\partial x}\right)^2 \cdot \frac{1}{2m}$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

1.3 Total energy for a bound particle: $E = \frac{p^2}{2m} + V(x)$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi$$

Time dependent Schrödinger equation

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)$$

Try $\psi(x, t) = \phi(x) \cdot \tau(t)$:

$$i\hbar \cdot \phi \cdot \frac{\partial T}{\partial t} = -\frac{\hbar^2}{2m} \cdot T \cdot \frac{\partial^2 \phi}{\partial x^2} + V(x) \cdot \phi \cdot T$$

$$\frac{i\hbar}{T} \frac{\partial T}{\partial t} = -\frac{\hbar^2}{2m} \frac{1}{\phi} \frac{\partial^2 \phi}{\partial x^2} + V(x)$$

Both sides must be constant $\forall x, t$. They are equal to the system's energy.

$$E = \frac{1}{T} i\hbar \frac{\partial T}{\partial t} \quad E = -\frac{\hbar^2}{2m} \frac{1}{\phi} \frac{\partial^2 \phi}{\partial x^2} + V(x)$$

$$E \int dt = i\hbar \int \frac{1}{T} dT$$

$$Et + C = i\hbar \ln T$$

$$-\frac{i}{\hbar} (Et + C) = \ln T$$

$$T = A e^{-\frac{i}{\hbar} Et}$$

$$\Psi(x, t) = \phi(x) \cdot A e^{-\frac{i}{\hbar} Et}$$

Note that $|\Psi(x, t)|^2 = |\phi(x)|^2$ (if A is absorbed).

$$E = -\frac{\hbar^2}{2m} \frac{1}{\phi} \frac{\partial^2 \phi}{\partial x^2} + V(x)$$

$$E\phi = -\frac{\hbar^2}{2m} \frac{\partial^2 \phi}{\partial x^2} + V(x)\phi$$

$$\hat{H}\phi = E\phi \quad (\text{Time independent Schrödinger equation})$$

This is an eigenvalue problem: the eigenvalues of $\hat{H}$ give the energies of the system.

1.4 $\int |\psi|^2 dx = 1$ This must hold because $|\psi|^2$ is a probability and cannot be > 1

$$\frac{d}{dt} \left(\int |\psi|^2 dx \right) = \frac{d}{dt} \left(\int \psi \psi^* dx \right)$$

$$= \int_{-\infty}^{\infty} \frac{\partial}{\partial t} (\psi \psi^*) dx$$

$$I = \int_{-\infty}^{\infty} \psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} dx$$

From TDSE: $i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$

$$\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} \hat{H} \psi$$

$$\frac{\partial \psi^*}{\partial t} = \frac{i}{\hbar} \hat{H} \psi^*$$

$$I = \int_{-\infty}^{\infty} -\frac{i}{\hbar} \psi^* \hat{H} \psi + \frac{i}{\hbar} \psi \hat{H} \psi^* dx$$

$$= \int_{-\infty}^{\infty} -\frac{i}{\hbar} \psi^* \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi \right) + \frac{i}{\hbar} \psi \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V(x) \psi^* \right) dx$$

$$= \int_{-\infty}^{\infty} \frac{i\hbar}{2m} \psi^* \frac{\partial^2 \psi}{\partial x^2} - \cancel{\frac{i}{\hbar} \psi^* V(x) \psi} - \cancel{\frac{i\hbar}{2m} \psi \frac{\partial^2 \psi^*}{\partial x^2}} + \cancel{\frac{i}{\hbar} \psi V(x) \psi^*} dx$$

$$= \frac{i\hbar}{2m} \int_{-\infty}^{\infty} \psi^* \frac{\partial^2 \psi}{\partial x^2} - \psi \frac{\partial^2 \psi^*}{\partial x^2} dx$$

$$= i\hbar \left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right]_{-\infty}^{\infty}$$

$$= 0$$

Since ψ was initially normalised, it cannot have a finite value at $\pm\infty$ (must be 0).

Part II: Boundary Conditions and the Infinite Square Well

2.1 $0 \leq x \leq L$ $V=0$ inside box, $V=\infty$ outside.

$$E\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi$$

$\psi=0$ outside the box because if $V(x)=\infty$, infinite energy would be required for that state (if ψ had a non zero value).

ψ must similarly equal 0 at $x=0$ and $x=L$ because it is continuous.

General solution: $\psi(x) = A \sin kx + B \cos kx$

$$\psi(0)=0 \Rightarrow B=0$$

$$\psi(L)=0 \Rightarrow A \sin kL = 0$$

$$kL = 0, \pi, 2\pi, \dots, n\pi$$

$$k = \frac{n\pi}{L} \quad n \in \mathbb{Z}$$

$$E_n = \frac{p_n^2}{2m}$$

$$p_n = \hbar k$$

$$= \frac{\hbar n\pi}{L}$$

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2}$$

Discretisation emerges purely from boundary conditions. Also note that $E_n \neq 0$. n is a quantum number.

$$\psi_n = A \sin\left(\frac{n\pi x}{L}\right) \quad \int_0^L |\psi_n|^2 dx = 1$$

$$A^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$= 1 - 2\sin^2\theta$$

$$\frac{1}{2}(1 - \cos 2\theta) = \sin^2\theta$$

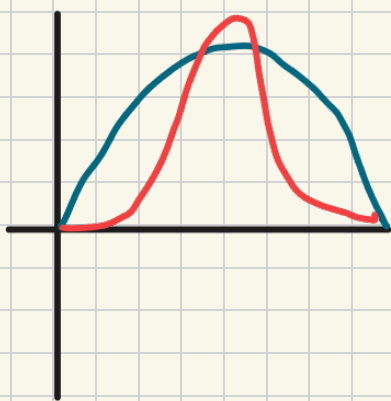
$$\frac{A^2}{2} \int_0^L 1 - \cos\left(\frac{2n\pi x}{L}\right) dx = 1$$

$$\frac{A^2}{2} \left[x - \frac{L}{2n\pi} \sin\left(\frac{2n\pi x}{L}\right) \right]_0^L = 1$$

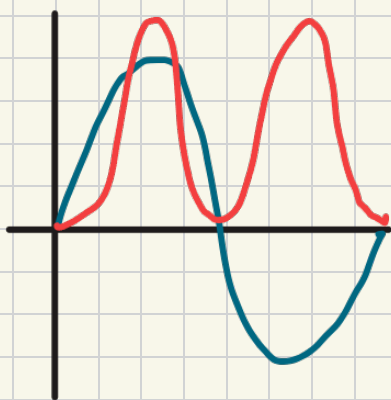
$$\frac{A^2}{2} [L - 0 - 0 + 0] = 1$$

$$A = \sqrt{\frac{2}{L}}$$

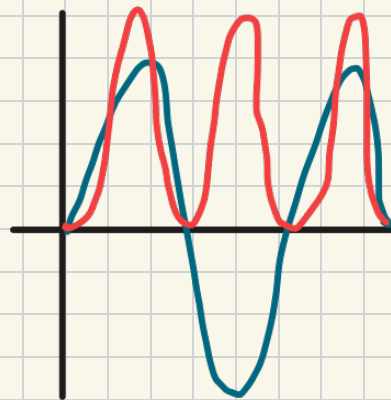
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$



$n=1$



$n=2$



$n=3$

2.2 The n th state has $n-1$ nodes. The analogous classical condition is standing waves on a string.

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2}$$

$$E_n \propto n^2$$

The spacing between energy levels increases as n becomes larger.

As $n \rightarrow \infty$, $|\psi_n|^2$ is nearly uniform, which implies that the particle is equally likely to be found anywhere.

$$\begin{aligned}
 2.3 \quad & \int_0^L \psi_m(x) \psi_n(x) dx \qquad \sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)] \\
 &= \int_0^L \frac{2}{L} \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \\
 &= \frac{1}{L} \int_0^L \cos\left(\frac{\pi x(n-m)}{L}\right) - \cos\left(\frac{\pi x(n+m)}{L}\right) dx \\
 &= \frac{1}{L} \left[\frac{L}{(n-m)\pi} \sin\left(\frac{\pi x(n-m)}{L}\right) - \frac{L}{(n+m)\pi} \sin\left(\frac{\pi x(n+m)}{L}\right) \right]_0^L \\
 &= 0 \quad (m \neq n)
 \end{aligned}$$

$$\therefore \int_0^L \psi_m(x) \psi_n(x) dx = \delta_{nm}$$

$$\psi(x, 0) = \sum_{n=1}^{\infty} c_n \psi_n(x)$$

$$\begin{aligned}
 \int_0^L \psi_m(x) \psi(x) dx &= \sum_{n=1}^{\infty} c_n \int_0^L \psi_m(x) \psi_n(x) dx \\
 &= \sum_{n=1}^{\infty} c_n \delta_{nm} \\
 &= c_m
 \end{aligned}$$

$|c_n|^2$ is the probability of measuring E_n . This is formally identical to a Fourier series because ψ_n form an orthonormal basis for functions on $[0, L]$ in the same way sines and

cosines are a basis in Fourier analysis.

Part III: The Harmonic Oscillator

$$3.1 \quad [\hbar] = M L^2 T^{-1} \quad [m] = M$$

$$[\omega] = T^{-1} \quad [x_0] = L$$

$$x_0 = \hbar^\alpha m^\beta \omega^\gamma$$

$$L = (M L^2 T^{-1})^\alpha M^\beta T^{-\gamma}$$

$$\alpha + \beta = 0$$

$$\beta = -\frac{1}{2}$$

$$2\alpha = 1$$

$$\alpha = \frac{1}{2}$$

$$\gamma = -\frac{1}{2}$$

$$-\alpha - \gamma = 0$$

$$x_0 = \sqrt{\frac{\hbar}{m\omega}}$$

Harmonic oscillator:

$$V(x) = \frac{1}{2} m \omega^2 x^2$$

$$= \frac{1}{2} x_0^2 m \omega^2 \xi^2$$

$\xi = \frac{x}{x_0}$ is a natural length scale.

$$\epsilon = \frac{2E}{\hbar\omega} \quad E = \frac{1}{2} \hbar\omega \epsilon$$

$$\text{TISE: } \frac{1}{2} \hbar\omega \epsilon \psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = \frac{1}{2} m \omega^2 x^2 \psi - \frac{1}{2} \hbar\omega \epsilon \psi$$

$$\frac{\partial \psi(\xi)}{\partial x} = \frac{\partial \psi}{\partial \xi} \cdot \frac{\partial \xi}{\partial x}$$

$$\frac{\partial \xi}{\partial x} = \frac{1}{x_0}$$

$$= \frac{1}{x_0} \cdot \frac{\partial \psi}{\partial \xi}$$

$$\begin{aligned}\frac{\partial^2 \psi^2}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{1}{x_0} \cdot \frac{\partial \psi}{\partial \xi} \right) \\ &= \frac{1}{x_0} \cdot \frac{1}{x_0} \frac{\partial^2 \psi}{\partial \xi^2} \\ &= \frac{1}{x_0^2} \frac{\partial^2 \psi}{\partial \xi^2}\end{aligned}$$

$$\text{TISE: } -\frac{\hbar^2}{2mx_0^2} \frac{\partial^2 \psi}{\partial \xi^2} = -\frac{1}{2} x_0^2 m \omega^2 \xi^2 \psi - E \psi$$

$$x_0^2 = \frac{\hbar}{m\omega}$$

$$\frac{-\cancel{\hbar^2} m \omega \frac{\partial^2 \psi}{\partial \xi^2}}{2 \cancel{\hbar}} = -\frac{\cancel{m \omega^2} \cdot \hbar}{2 \cancel{m \omega}} \xi^2 \psi - E \psi$$

$$-\frac{\hbar \omega}{2} \frac{\partial^2 \psi}{\partial \xi^2} = -\frac{\hbar \omega}{2} \xi^2 \psi - E \psi$$

$$E \psi = -\frac{\hbar \omega}{2} \frac{\partial^2 \psi}{\partial \xi^2} + \frac{\hbar \omega}{2} \xi^2 \psi$$

$$\frac{2E}{\hbar \omega} \psi = -\frac{\partial^2 \psi}{\partial \xi^2} + \xi^2 \psi$$

$$\frac{\partial^2 \psi}{\partial \xi^2} = (\xi^2 - \epsilon) \psi$$

As ξ tends to infinity, $\xi^2 \gg \epsilon$, so the equation simplifies to:

$$\frac{\partial^2 \psi}{\partial \xi^2} - \xi^2 \psi = 0$$

$$\text{Trial: } \psi = A e^{\alpha \xi^2} \quad \psi' = 2A \alpha \xi e^{\alpha \xi^2}$$

$$\psi'' = 2\alpha e^{\alpha \xi^2} + 4\alpha^2 \xi^2 e^{\alpha \xi^2}$$

$$= 2\alpha e^{\alpha \xi^2} (1 + 2\alpha \xi^2)$$

As $\xi \rightarrow \infty$, $(1 + 2\alpha \xi^2) \rightarrow 2\xi^2$

$$\psi'' \rightarrow 4\alpha^2 \xi^2 e^{\alpha \xi^2}$$

~~$$4\alpha^2 \xi^2 e^{\alpha \xi^2} = \xi^2 e^{\alpha \xi^2}$$~~

$$4\alpha^2 = 1$$

$$\alpha = \pm \frac{1}{2}$$

Positive solution rejected, as $\psi \rightarrow 0$ as $\xi \rightarrow \infty$ (must be normalisable).

$$\therefore \psi = H(\xi) e^{-\frac{1}{2}\xi^2}$$

3.2 Since we do not know what H is, use power series representation $H(\xi) = \sum a_j \xi^j$.

Recall that: $\frac{\partial^2 \psi}{\partial \xi^2} = (\xi^2 - \epsilon) \psi$

$$H'(\xi) = \sum a_j j \xi^{j-1}$$

$$H''(\xi) = \sum a_j j(j-1) \xi^{j-2}$$

$$= \sum a_{j+2} (j+2)(j+1) \xi^j$$

$$\psi(\xi) = \left(\sum a_j \xi^j \right) e^{-\frac{1}{2}\xi^2}$$

$$\psi' = H'(\xi) e^{-\frac{1}{2}\xi^2} - \xi H(\xi) e^{-\frac{1}{2}\xi^2}$$

$$\psi'' = H''(\xi) e^{-\frac{1}{2}\xi^2} - \xi H(\xi)' e^{-\frac{1}{2}\xi^2} - H(\xi) e^{-\frac{1}{2}\xi^2}$$

$$- \xi H'(\xi) e^{-\frac{1}{2}\xi^2} + \xi^2 H(\xi) e^{-\frac{1}{2}\xi^2}$$

$$H'' - 2\xi H' + (\cancel{\xi^2} - 1)H = (\cancel{\xi^2} - \epsilon)H$$

$$H'' - 2\xi H' + (\epsilon - 1)H = 0$$

$$\sum a_{j+2} (j+2)(j+1) \xi^j - 2 \sum a_j j \xi^j + (\epsilon - 1) \sum a_j \xi^j = 0$$

$$\sum \xi^j [a_{j+2} (j+2)(j+1) - 2a_j j + \epsilon a_j - a_j] = 0$$

$$a_{j+2} (j+2)(j+1) - 2a_j j + \epsilon a_j - a_j = 0$$

$$a_{j+2} (j+2)(j+1) = a_j (2j - \epsilon + 1)$$

$$a_{j+2} = \frac{2j - \epsilon + 1}{(j+2)(j+1)} a_j$$

For the behaviour of the series if it does not terminate, we consider very large j :

$$\frac{a_{j+2}}{a_j} = \frac{2j - \epsilon + 1}{(j+2)(j+1)}$$

$$\approx \frac{2}{j}$$

as $j \rightarrow \infty$

Consider the Taylor expansion of e^{ξ^2} :

$$e^{\xi^2} \approx \sum_{j=0}^{\infty} \frac{\xi^{2j}}{j!}$$

$$b_j = \frac{1}{(j/2)!}$$

$$b_{j+2} = \frac{1}{(j/2+1)!}$$

$$\frac{b_{j+2}}{b_j} = \frac{1}{\frac{j}{2} + 1} = \frac{2}{j+2} \approx \frac{2}{j} \text{ for large } j$$

So if the series does not terminate, the function it

approximates is of the form e^{ξ^2} . This makes ψ not normalisable, which is unphysical. So $H(\xi)$ must terminate. It terminates if $a_{j+2} = 0$:

$$2n - \epsilon + 1 = 0$$

$$\epsilon = 2n + 1$$

$$E_n = \frac{1}{2} \hbar \omega (2n + 1)$$

$$= (n + \frac{1}{2}) \hbar \omega$$

$E_0 = \frac{\hbar \omega}{2}$ is the ground state, which is the lowest possible energy state. Interestingly, it is not equal to 0, which is a consequence of the uncertainty principle.

3.3 Define two ladder operators:

$$\hat{a}_+ = \frac{1}{\sqrt{2}} \left(\xi - \frac{d}{d\xi} \right) \quad \hat{a}_- = \frac{1}{\sqrt{2}} \left(\xi + \frac{d}{d\xi} \right)$$

$$\hat{a}_+ \hat{a}_- = \frac{1}{2} \left(\xi - \frac{d}{d\xi} \right) \left(\xi + \frac{d}{d\xi} \right)$$

$$= \frac{1}{2} \left(\xi^2 f + \xi f' - \frac{d}{d\xi} (\xi f) - f'' \right)$$

$$= \frac{1}{2} \left[\xi^2 + \cancel{\xi f'} - \left(f \frac{d}{d\xi} (\xi) + \cancel{\xi \frac{d}{d\xi} f} \right) - f'' \right]$$

$$= \frac{1}{2} \left(\xi^2 - \frac{d^2}{d\xi^2} - 1 \right)$$

$$\hat{H} = \frac{\hbar \omega}{2} \left(\xi^2 - \frac{d^2}{d\xi^2} \right)$$

$$\therefore \hat{H} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)$$

$$[\hat{a}_-, \hat{a}_+] = \hat{a}_- \hat{a}_+ - \hat{a}_+ \hat{a}_-$$

$$\begin{aligned} \hat{a}_- \hat{a}_+ &= \hat{a}_+ \hat{a}_- + 1 \\ \hat{a}_+ \hat{a}_- &= \hat{a}_- \hat{a}_+ - 1 \end{aligned}$$

$$\hat{a}_- \hat{a}_+ = \frac{1}{2} \left(\xi + \frac{d}{d\xi} \right) \left(\xi f - f' \right)$$

$$= \frac{1}{2} \left(\xi^2 f - \cancel{\xi f'} + f + \cancel{\xi f'} - f'' \right)$$

$$= \frac{1}{2} \left(\xi^2 - \frac{d^2}{d\xi^2} + 1 \right)$$

$$[\hat{a}_-, \hat{a}_+] = \frac{1}{2} \left(\cancel{\xi^2} - \cancel{\frac{d^2}{d\xi^2}} + 1 \right) - \frac{1}{2} \left(\cancel{\xi^2} - \cancel{\frac{d^2}{d\xi^2}} - 1 \right)$$

$$= 1$$

Now consider: $\hat{H}\psi = E\psi$

$$\hbar\omega \left(\hat{a}_+ \hat{a}_- \psi + \frac{1}{2} \psi \right) = E\psi$$

$$\hat{H}(\hat{a}_+ \psi) = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) (\hat{a}_+ \psi)$$

$$= \hbar\omega \left(\hat{a}_+ \hat{a}_- \hat{a}_+ \psi + \frac{1}{2} \hat{a}_+ \psi \right)$$

$$= \hbar\omega \left[\hat{a}_+ (\hat{a}_+ \hat{a}_- + 1) \psi + \frac{1}{2} \hat{a}_+ \psi \right]$$

$$= \hbar\omega \left(\hat{a}_+ \hat{a}_+ \hat{a}_- \psi + \hat{a}_+ \psi + \frac{1}{2} \hat{a}_+ \psi \right)$$

$$= \hbar\omega \hat{a}_+ \left(\hat{a}_+ \hat{a}_- \psi + \psi + \frac{1}{2} \psi \right)$$

$$= \hat{a}_+ \left[\hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) \psi + \hbar\omega \psi \right]$$

$$= (E + \hbar\omega) \hat{a}_+ \psi$$

So $\hat{a}_+ \psi$ is an eigenstate with energy $E + \hbar\omega$. Similarly, $\hat{a}_- \psi$ is an eigenstate with energy $E - \hbar\omega$.

Since energies must be positive, there must be a ground state:

$$\hat{a}_- \psi_0 = 0$$

$$\frac{1}{\sqrt{2}} \left(\xi + \frac{d}{d\xi} \right) \psi_0 = 0$$

$$\frac{1}{\sqrt{2}} \psi_0 \xi + \frac{1}{\sqrt{2}} \frac{d\psi_0}{d\xi} = 0$$

$$\frac{d\psi_0}{d\xi} = -\psi_0 \xi$$

$$\int \frac{1}{\psi_0} d\psi_0 = -\int \xi d\xi$$

$$\psi_0 = A e^{-\frac{\xi^2}{2}}$$

$$\hat{H} \psi_0 = \hbar\omega \left(\cancel{\hat{a}_+ \hat{a}_-} \psi_0 + \frac{1}{2} \psi_0 \right)$$

$$E \psi_0 = \frac{\hbar\omega}{2} \psi_0$$

$$E_0 = \frac{\hbar\omega}{2}$$

$$E_n = \left(n + \frac{1}{2} \right) \hbar\omega$$

which is the same result as the series ODE result, but with a simpler method.

Part IV: The Hydrogen Atom

4.1 $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

We want to use spherical coordinates because the potential is spherically symmetrical.

$$r^2 = x^2 + y^2 + z^2$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

Consider that $\psi(r, \theta, \phi) = R(r) \cdot Y(\theta, \phi)$.

TISE: $\hat{H}\psi = E\psi$

$$\hat{H}\psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(r)\psi$$

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

Radial part:

$$\begin{aligned} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 Y(\theta, \phi) \frac{dR}{dr} \right) \\ &= Y \cdot \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) \quad \leftarrow \text{Radial} \end{aligned}$$

Angular part:

$$\begin{aligned} \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} \\ = \frac{R}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dY}{d\theta} \right) + \frac{R}{r^2 \sin^2 \theta} \frac{d^2 Y}{d\phi^2} \quad \leftarrow \text{Angular} \end{aligned}$$

For the kinetic part of the TISE, the above two parts are added and multiplied by $-\frac{\hbar^2}{2m}$. To simplify and separate,

we multiply by $-\frac{2mr^2}{\hbar^2}$ and divide by RY .

Kinetic term:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{Y} \cdot \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dY}{d\theta} \right) + \frac{1}{Y} \cdot \frac{1}{\sin^2 \theta} \frac{d^2 Y}{d\phi^2}$$

Potential term:

$$V(r) RY \cdot \frac{1}{RY} \cdot -\frac{2mr^2}{\hbar^2} = -\frac{2mr^2}{\hbar^2} V(r)$$

Total energy term:

$$E \cdot RY \cdot \frac{1}{RY} \cdot -\frac{2mr^2}{\hbar^2} = -\frac{2mr^2}{\hbar^2} E$$

Separation:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} (V(r) - E) = -\frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dY}{d\theta} \right) + \frac{1}{\sin^2 \theta} \frac{d^2 Y}{d\phi^2} \right]$$

One side depends only on r and one side depends only on θ and ϕ . However, since they equal each other, they must both be constant. Call this constant $l(l+1)$.

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} (V(r) - E) = l(l+1)$$

$$-\frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dY}{d\theta} \right) + \frac{1}{\sin^2 \theta} \frac{d^2 Y}{d\phi^2} \right] = l(l+1)$$

4.2 Let $Y(\theta, \phi) = \Theta(\theta) \Phi(\phi)$.

$$-\frac{1}{\Theta \Phi} \left[\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \Phi \frac{d\Theta}{d\theta} \right) + \frac{\Theta}{\sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} \right] = l(l+1)$$

$$-\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + l(l+1) \sin^2 \theta$$

Φ equation:

$$-\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = m^2$$

$$\frac{d^2 \Phi}{d\phi^2} + m^2 \Phi = 0$$

AQE: $\lambda^2 + m = 0$

$$\lambda = \pm im$$

GS: $\Phi = A e^{im\phi}$

$$A e^{im\phi} = A e^{im(\phi+2\pi)}$$

$$= A e^{im\phi + 2\pi im}$$

$$= A e^{im\phi} e^{2\pi im}$$

Since ϕ represents azimuthal angle,

$$\Phi(\phi) = \Phi(\phi + 2\pi).$$

$$e^{2\pi im} = 1$$

$$\cos(2\pi m) = 1 \quad m = 0, 1, 2, \dots, n$$

$\therefore m$ must be an integer

Θ equation:

$$m^2 = \frac{\sin\theta}{\Theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) + l(l+1) \sin^2\theta$$

$$\frac{1}{\Theta} \cdot \frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) + \left(l(l+1) - \frac{m^2}{\sin^2\theta} \right) = 0$$

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) + \left(l(l+1) - \frac{m^2}{\sin^2\theta} \right) \Theta = 0$$

Let $u = \cos\theta$

$$\frac{du}{d\theta} = -\sin\theta \quad \sin^2\theta = 1 - u^2$$

$$\frac{d\Theta}{d\theta} = \frac{d\Theta}{du} \cdot \frac{du}{d\theta}$$

$$= -\sin\theta \frac{d\Theta}{du}$$

$$\frac{d}{d\theta} = -\sin\theta \frac{d}{du}$$

First term:

$$\begin{aligned}\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \cdot -\sin \theta \frac{d\Theta}{dv} \right) &= \frac{1}{\sin \theta} \frac{d}{d\theta} \left[(v^2 - 1) \frac{d\Theta}{dv} \right] \\ &= -\frac{d}{dv} \left[(v^2 - 1) \frac{d\Theta}{dv} \right] \\ &= (1 - v^2) \frac{d^2 \Theta}{dv^2} - 2v \frac{d\Theta}{dv}\end{aligned}$$

Second term:

$$\left(l(l+1) - \frac{m^2}{\sin^2 \theta} \right) \Theta = \left(l(l+1) - \frac{m^2}{1-v^2} \right) \Theta$$

$$(1-v^2) \frac{d^2 \Theta}{dv^2} - 2v \frac{d\Theta}{dv} + \left(l(l+1) - \frac{m^2}{1-v^2} \right) \Theta = 0$$

Try $\Theta(v) = (1-v^2)^{\frac{m}{2}} y(v)$ substitution:

$$f(v) = -m(1-v^2)^{\frac{m}{2}-1} + m(m-2)v(1-v^2)^{\frac{m}{2}-2}$$

$$\Theta' = y' (1-v^2)^{\frac{m}{2}} - y m v (1-v^2)^{\frac{m}{2}-1}$$

$$\begin{aligned}\Theta'' &= y'' (1-v^2)^{\frac{m}{2}} - y' m v (1-v^2)^{\frac{m}{2}-1} - y \frac{d}{dv} [m v (1-v^2)^{\frac{m}{2}-1}] - \\ &\quad y' m v (1-v^2)^{\frac{m}{2}-1}\end{aligned}$$

$$\Theta' = (1-v^2)^{\frac{m}{2}} y'' - 2m v (1-v^2)^{\frac{m}{2}-1} y' - f(v) y$$

Set $g = (1-v^2)^{\frac{m}{2}}$.

$$\Theta = g y \quad \Theta' = g' y + y' g \quad \Theta'' = g'' y + 2g' y' + g y''$$

First term:

$$(1-v^2) g'' y + 2(1-v^2) g' y' + (1-v^2) g y''$$

Second term:

$$-2v g' y - 2v y' g$$

Third term:

$$\left[l(l+1) - \frac{m^2}{1-v^2} \right] g y$$

$$g' = -mv(1-v^2)^{\frac{m}{2}-1}$$

$$g'' = m(1-v^2)^{\frac{m}{2}-1} + m(m-2)v^2(1-v^2)^{\frac{m}{2}-2}$$

Expand and divide all terms by g :

First term:

$$-my + \frac{m(m-2)v^2}{1-v^2}y - 2mv y' + (1-v^2)y''$$

Second term:

$$\frac{2mv^2}{1-v^2}y - 2v y'$$

Third term:

$$l(l+1)y - \frac{m^2}{1-v^2}y$$

$$y'' \text{ terms: } (1-v^2)$$

$$y' \text{ terms: } -2(m+1)v$$

$$y \text{ terms: } -m + \frac{m^2-2m}{1-v^2}v^2 + \frac{2mv^2}{1-v^2} - \frac{m^2}{1-v^2} + l(l+1)$$

$$= \frac{m^2v^2 - \cancel{2mv^2} + \cancel{2mv^2} - m^2}{1-v^2} - m + l(l+1)$$

$$= \frac{m^2(v^2-1)}{1-v^2} - m + l(l+1)$$

$$= -m(m+1) + l(l+1)$$

ODE:

$$(1-v^2)y'' - 2(m+1)v y' + [l(l+1) - m(m+1)]y = 0$$

Assume the form $y = \sum_{n=0}^{\infty} a_n v^n$

$$y' = \sum_{n=1}^{\infty} n a_n v^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n v^{n-2}$$

$$(1-v^2) \sum_{n=2}^{\infty} n(n-1) a_n v^{n-2} - 2(m+1)v \sum_{n=1}^{\infty} n a_n v^{n-1} + [l(l+1) - m(m+1)] \sum_{n=0}^{\infty} a_n v^n = 0$$

First term:

$$\sum_{n=2}^{\infty} n(n-1) a_n v^{n-2} - \sum_{n=2}^{\infty} n(n-1) a_n v^n \quad k=n-2 \text{ (dummy index)}$$

$$= \sum_{k=2}^{\infty} (k+2)(k+1) a_{k+2} v^k - \sum_{n=2}^{\infty} n(n-1) a_n v^n$$

$$= \sum_{n=0}^{\infty} v^n [(n+2)(n+1) a_{n+2} - n(n-1) a_n]$$

Second term:

$$\sum_{n=0}^{\infty} -2(m+1)n a_n v^n$$

Third term:

$$\sum_{n=0}^{\infty} [l(l+1) - m(m+1)] a_n v^n$$

Combine:

$$\sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} - n(n-1) a_n - 2(m+1)n a_n + [l(l+1) - m(m+1)] a_n] v^n = 0$$

$$a_{n+2} (n+2)(n+1) = a_n (l(l+1) - m(m+1) - n(n-1) - 2n(m+1))$$

$$= a_n (l^2 + l - m^2 - m - n^2 + n - 2(m+1)n)$$

$$= a_n [l(l+1) - (n+m)(n+m+1)]$$

$$a_{n+2} = \frac{l(l+1) - (n+m)(n+m+1)}{(n+2)(n+1)} a_n$$

Note that the above produces an even series solution $P(n)$ and an odd series solution $Q(n)$.

The series terminates for $l(l+1) = (n+m)(n+m+1)$. It must

terminate because it forms part of the wavefunction so cannot diverge.

$$l(l+1) = (n+m)(n+m+1)$$

$$n+m = l$$

$$n = l - m$$

$$l \geq 0 \quad m \leq |l| \leq n \text{ must be +ve.}$$

$$\therefore \Theta = P_l^m(u)$$

From termination conditions and necessity of non-zero polynomial.

$$Y_l^m(\theta, \phi) = P_l^m(\cos\theta) e^{im\phi}$$

4.3 The radial equation is:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} (V(r) - E) = l(l+1)$$

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \underbrace{\frac{2me^2 r}{4\pi\epsilon_0 \hbar^2} + \frac{2mEr^2}{\hbar^2}} = l(l+1)$$

Quadratic in r which $\rightarrow 0$ as $r \rightarrow 0$. So as $r \rightarrow 0$, $\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) \rightarrow l(l+1)$. Assume $R(r) \approx r^s$:

$$r^{-s} \frac{d}{dr} \left[r^2 \frac{d}{dr} (r^s) \right] = r^{-s} \frac{d}{dr} (r^2 \cdot s r^{s-1})$$

$$= r^{-s} \frac{d}{dr} (s r^{s+1})$$

$$= r^{-s} [s(s+1) r^s]$$

$$= s(s+1) \quad \therefore s = l$$

$$\text{So } R(r) \approx r^l.$$

As $r \rightarrow \infty$, $V(r) \rightarrow 0$:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2mEr^2}{\hbar^2} = l(l+1)$$

$$l(l+1) \ll \left| \frac{2mEr^2}{\hbar^2} \right|$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \frac{2mEr^2}{\hbar^2}$$

$E < 0$ for bound states.

Try $R \approx e^{dr}$:

$$e^{-dr} \frac{d}{dr} (dr^2 e^{dr}) = \frac{2mEr^2}{\hbar^2}$$

$$\cancel{e^{-dr}} (2\cancel{dr} \cancel{e^{dr}} + d^2 r^2 \cancel{e^{dr}}) = \frac{2mEr^2}{\hbar^2}$$

$$d^2 r^2 + 2dr = \frac{2mE}{\hbar^2} r^2$$

As $r \rightarrow \infty$, $r^2 \gg r$.

$$\cancel{d^2 r^2} = \frac{2mE}{\hbar^2} \cancel{r^2}$$

$$d = - \left(\frac{2mE}{\hbar^2} \right)^{\frac{1}{2}}$$

$R \rightarrow 0$ as $r \rightarrow \infty$
because it needs to be normalisable.

$$\therefore R(r) = r^l \cdot e^{-\frac{r}{a}} \cdot v(r) \quad \text{where } a = \frac{\hbar}{\sqrt{2m|E|}} \quad \text{and } v(r)$$

$$v(r) = \sum_{k=0}^{\infty} c_k r^k$$

is a power series.

$$\frac{dR}{dr} = l r^{l-1} \left(e^{-\frac{r}{a}} \sum_{k=0}^{\infty} c_k r^k \right) - \frac{1}{a} e^{-\frac{r}{a}} \left(r^l \sum_{k=0}^{\infty} c_k r^k \right) + \sum_{k=0}^{\infty} k c_k r^{k-1} \left(r^l e^{-\frac{r}{a}} \right)$$

$$= r^{l-1} e^{-\frac{r}{a}} \left[l v - \frac{r v}{a} + r v' \right]$$

$$r^2 \frac{dR}{dr} = r^{l+1} e^{-\frac{r}{a}} \left(l v - \frac{r v}{a} + r v' \right)$$

$$\frac{1}{R} = \frac{1}{r^l e^{-\frac{r}{a}} v}$$

$$\begin{aligned} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) &= (l+1) r^l e^{-\frac{r}{a}} \left(l v - \frac{r v}{a} + r v' \right) - \frac{1}{a} r^{l+1} e^{-\frac{r}{a}} \left(l v - \frac{r v}{a} + r v' \right) \\ &\quad + \left(l v' - \frac{r v'}{a} - \frac{v}{a} + r v'' + v' \right) r^{l+1} e^{-\frac{r}{a}} \\ &= r^l e^{-\frac{r}{a}} \left[(l+1) \left(l v - \frac{r v}{a} + r v' \right) - \left(\frac{r}{a} \right) \left(l v - \frac{r v}{a} + r v' \right) \right. \\ &\quad \left. + r \left(l v' - \frac{r v'}{a} - \frac{v}{a} + r v'' + v' \right) \right] \end{aligned}$$

Dividing by $r^l e^{-\frac{r}{a}}$:

$$\begin{aligned} &= l^2 v - \frac{l r v}{a} + l r v' + l v - \frac{r v}{a} + r v' - \frac{l r v}{a} + \frac{r^2 v}{a^2} - \frac{r^2 v'}{a} + r l v' - \frac{r^2 v'}{a} \\ &\quad - \frac{r v}{a} + r^2 v'' + r v' \end{aligned}$$

$$= r^2 v'' + v' \left(l r + r - \frac{r^2}{a} + l r - \frac{r^2}{a} + r \right) + v \left(l^2 - \frac{l r}{a} + l - \frac{r}{a} - \frac{l r}{a} + \frac{r^2}{a^2} - \frac{r}{a} \right)$$

$$= r^2 v'' + 2 r v' \left(l + 1 - \frac{r}{a} \right) + v \left(l^2 + l - \frac{2 l r + 2 r}{a} + \frac{r^2}{a^2} \right)$$

$$= r^2 v'' + 2 r v' \left(l + 1 - \frac{r}{a} \right) + v \left[l(l+1) - \frac{2 r(l+1)}{a} + \frac{r^2}{a^2} \right]$$

$$\therefore \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \frac{1}{v} \left(r^2 v'' + 2 r v' \left(l + 1 - \frac{r}{a} \right) + v \left[l(l+1) - \frac{2 r(l+1)}{a} + \frac{r^2}{a^2} \right] \right)$$

$$v(r) = \sum_{k=0}^{\infty} c_k r^k \quad v'(r) = \sum_{k=0}^{\infty} k c_k r^{k-1} \quad v''(r) = \sum_{k=0}^{\infty} k(k-1) c_k r^{k-2}$$

$$\begin{aligned} &r^2 v'' + 2 r v' \left(l + 1 - \frac{r}{a} \right) + v \left[\cancel{l(l+1)} - \frac{2 r(l+1)}{a} + \frac{r^2}{a^2} \right] + v \left(\frac{2 m r}{4 \pi \epsilon_0 \hbar^2} + \frac{2 m E r^2}{\hbar^2} \right) \\ &= \cancel{v l(l+1)} \end{aligned}$$

Potential terms: $v\left(\frac{2me^2r}{4\pi\epsilon_0\hbar^2} + \frac{2mEr^2}{\hbar^2}\right)$

$$r^2 v'' + 2rv'\left(l+1 - \frac{r}{a}\right) + v\left(-\frac{2r(l+1)}{a} + \frac{r^2}{a^2} + \frac{2mEr^2}{\hbar^2} + \frac{2me^2r}{4\pi\epsilon_0\hbar^2}\right) = 0$$

v'' term: $\sum_{k=0}^{\infty} k(k-1)c_k r^k$

$$\begin{aligned} v' \text{ term: } \sum_{k=0}^{\infty} 2kc_k r^k \left(l+1 - \frac{r}{a}\right) &= \sum_{k=0}^{\infty} 2k(l+1)c_k r^k - \sum_{k=0}^{\infty} \frac{2kc_k}{a} r^{k+1} \\ &= \sum_{k=0}^{\infty} 2k(l+1)c_k r^k - \sum_{k=1}^{\infty} \frac{2(k-1)}{a} c_{k-1} r^k \\ &= \sum_{k=0}^{\infty} \left[2k(l+1)c_k - \frac{2(k-1)}{a} c_{k-1}\right] r^k \end{aligned}$$

$$\begin{aligned} v \text{ term: } \sum_{k=0}^{\infty} c_k r^{k+1} \left(-\frac{2(l+1)}{a} + \frac{2me^2}{4\pi\epsilon_0\hbar^2}\right) &+ \sum_{k=0}^{\infty} c_k r^{k+2} \left(\frac{1}{a^2} + \frac{2mE}{\hbar^2}\right) \\ &= \sum_{k=1}^{\infty} c_{k-1} r^k \left(-\frac{2(l+1)}{a} + \frac{2me^2}{4\pi\epsilon_0\hbar^2}\right) + \sum_{k=2}^{\infty} c_{k-2} r^k \left(\frac{1}{a^2} + \frac{2mE}{\hbar^2}\right) \end{aligned}$$

Combined: $\sum_{k=0}^{\infty} (\text{combined terms}) r^k = 0$

(combined terms) = 0

$$c_k \left[k(k-1) + 2k(l+1) \right] + c_{k-1} \left[-\frac{2(k+1)}{a} + \frac{2me^2}{4\pi\epsilon_0\hbar^2} \right] + c_{k-2} \left(\frac{1}{a^2} + \frac{2mE}{\hbar^2} \right) = 0$$

$\quad \quad \quad = A \quad \quad \quad \quad \quad = B \quad \quad \quad \quad \quad = C$

$$c_k = -\frac{B}{A} c_{k-1} - \frac{C}{A} c_{k-2}$$

The series is infinite if it does not terminate. Test for divergence.

$$c_{k+1} = -\frac{B}{A}c_k - \frac{C}{A}c_{k-1}$$

$$\frac{c_{k+1}}{c_k} = -\frac{B}{A} - \frac{C}{A} \frac{c_{k-1}}{c_k} \quad L = \lim_{k \rightarrow \infty} \left| \frac{c_{k+1}}{c_k} \right|$$

$$\frac{c_{k+1}}{c_k} = \frac{c_k}{c_{k-1}} = \frac{1}{L} \quad L = -\frac{B}{A} - \frac{C}{A} \cdot \frac{1}{L}$$

$$L^2 + \frac{B}{A}L - \frac{C}{A} = 0$$

$$AL^2 + BL - C = 0$$

$$L = \frac{-B \pm \sqrt{B^2 + 4AC}}{2A}$$

To show that it diverges, we need to show that $L \neq 0$.
 $L = 0 \Leftrightarrow C = 0$:

$$\frac{1}{a^2} + \frac{2mE}{\hbar^2} = C$$

$$\frac{2mE}{\hbar^2} = -\frac{2m|E|}{\hbar^2}$$

$$\frac{1}{a^2} = \frac{2m|E|}{\hbar^2}$$

$$C = 0$$

$$\Rightarrow L^2 + \frac{B}{A}L = 0$$

The root with the largest magnitude dominates at large k . So we need to show $\frac{B}{A} \neq 0 \Leftrightarrow B \neq 0$.

$$B = -\frac{2(k+1)}{a} + \frac{2me^2}{4\pi\epsilon_0\hbar^2}$$

$$\frac{1}{a} = \frac{\sqrt{2m|E|}}{\hbar}$$

$$B = \frac{-8\pi\epsilon_0 \hbar(k+l)(2m|E|)^{\frac{1}{2}} + 2me^2}{4\pi\epsilon_0 \hbar^2}$$

The two terms in the numerator do not collapse to 0. Therefore $B \neq 0$, $|\frac{B}{A}| > 0$, $L = -\frac{B}{A}$ and the series diverges for some choice of r . Since the solution must be normalisable, the series must terminate. Note that we have shown that $C=0$. So:

$$C_k = -\frac{B}{A} C_{k-1}$$

Terminates at some $k=n_{\max}$ where $C_{n_{\max}} \neq 0$ and $C_{n_{\max}+1} = 0$.

$$\cancel{-\frac{l(l+1)}{a}} = \cancel{\frac{2me^2}{4\pi\epsilon_0 \hbar^2}}$$

From $C=0$:

$$a = \frac{\hbar^2}{(2m|E|)^{\frac{1}{2}}}$$

$$k = n_{\max} + 1$$

$$a = \frac{4\pi\epsilon_0 \hbar^2 (n_{\max} + l + 1)}{me^2}$$

$$n = n_{\max} + l + 1$$

$$\frac{\hbar^2}{2\cancel{m}|E|} = \frac{16\pi^2 \epsilon_0^2 \hbar^4}{m^2 e^4} n^2$$

$$|E| = \frac{me^4}{32\pi^2 \epsilon_0^2 \hbar^2 n^2}$$

$$= -\frac{13.6}{n^2} \text{ eV}$$

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2}$$

= BOHR RADIUS

$$a = n a_0$$

$n_{\max} > 0$ (summation index):

$$n - l - 1 > 0$$

$$n > l + 1$$

So far the constraints on the quantum numbers n, l, m are:

$$n, m, l \in \mathbb{Z} \quad n \geq l+1 \quad l \geq 0 \quad m \leq |l| \quad n > 0$$

4.4

n	l	m	
1	0	0	$l=0: s$
2	0, 1	0, ± 1	$l=1: p$
3	0, 1, 2	0, $\pm 1, \pm 2$	$l=2: d$
			$l=3: f$

n determines energy levels. l determines the magnitude of orbital angular momentum. m determines the z -component of angular momentum so acts to orient it.

$$\psi(r, \theta, \phi) = R(r) \cdot Y_l^m(\theta, \phi)$$

$$P(r) dr = r^2 |R(r)|^2 dr$$

$$R(r) = r^l e^{-\frac{r}{a}} v(r)$$

1s: $n=1, l=0$

$$n_{\max} = 0 \quad v(r) = c_0 \quad a = n \cdot a_0 = a_0$$

$$R(r) = r^l e^{-r/a_0} c_0 = c_0 e^{-r/a_0}$$

$$P(r) = |c_0|^2 r^2 e^{-2r/a_0}$$

2s: $n=2, l=0$ $v(r) = c_0 + c_1 r$ $c_1 = -\frac{B}{A} c_0 \quad (k=1)$

$$n_{\max} = 1 \quad a = 2a_0$$

$$c_1 = -\frac{1}{2a_0} c_0$$

$$v(r) = c_0 \left(1 - \frac{r}{2a_0}\right)$$

$$A=2, \quad B = -\frac{(k+l)}{a_0} + \frac{2}{a_0} = \frac{1}{a_0}$$

$$R(r) = c_0 \left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0}$$

$$= c_0 \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$$

$$P(r) = r^2 |R(r)|^2$$

$$= r^2 |c_0|^2 \left(2 - \frac{r}{a_0}\right)^2 e^{-r/a_0}$$

$$2p: n=2, l=1 \quad v(r) = c_0 \quad a = 2a_0$$

$$R(r) = c_0 r e^{-r/2a_0}$$

$$P(r) = |c_0|^2 r^4 e^{-r/a_0}$$

For the 1s state, the most probable radius is found where

$$\frac{dP}{dr} = 0:$$

$$\frac{dP}{dr} = 2r |c_0|^2 e^{-2r/a_0} - \frac{2}{a_0} |c_0|^2 r^2 e^{-2r/a_0}$$

~~$$\frac{2}{a_0} |c_0|^2 r^2 e^{-2r/a_0} = 2r |c_0|^2 e^{-2r/a_0}$$~~

$$r = a_0$$

Part V: Numerical Solver

Recall the 1D TISE: $E\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi$

Divide the interval $a \leq x \leq b$ (boundary conditions $\psi(a) = \alpha$, $\psi(b) = \beta$) into N equal pieces: $h = \frac{b-a}{N}$. Grid points are defined as $x_i = a + ih$, $i = 0, 1, \dots, N$. The Taylor expansions for the forward and backward points $x_i + h$ and $x_i - h$ are found by first finding the Taylor expansion at a

general point x_i :

$$\psi(x) = \psi(x_i) + \psi'(x_i)(x - x_i) + \frac{\psi''(x_i)}{2!}(x - x_i)^2 + \dots$$

For the forward and backward points:

$$x = x_i + h \quad \text{and} \quad x = x_i - h$$

$$x - x_i = h \quad \quad \quad x - x_i = -h$$

$$\psi(x_i + h) = \psi(x_i) + h\psi'(x_i) + \frac{h^2}{2}\psi''(x_i) + \frac{h^3}{6}\psi'''(x_i) + \dots$$

$$\psi(x_i - h) = \psi(x_i) - h\psi'(x_i) + \frac{h^2}{2}\psi''(x_i) - \frac{h^3}{6}\psi'''(x_i) + \dots$$

Adding them gives:

$$\psi_{i-1} + \psi_{i+1} = 2\psi_i + h^2\psi''_i + \frac{h^4}{12}\psi^{(4)}_i + \dots$$

$$\psi''_i = \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2} + O(h^2)$$

For the TISE:

$$-\frac{\hbar^2}{2m} \left(\frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2} \right) + V(x)\psi_i = E\psi_i$$

First, this is calculated for the infinite square well, where $0 \leq x \leq L$ $V=0$ inside box, $V=\infty$ outside. At the boundaries, $\psi=0$ because of the normalisation condition.

$$\psi_0 = 0 \quad \psi_n = 0$$

$$-\frac{\hbar^2}{2m} \left(\frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2} \right) = E\psi_i$$

The coefficients of ψ_{i-1} , ψ_i and ψ_{i+1} are the lower, main and upper diagonals of the Hamiltonian matrix.

Every other element is 0. So:

$$\hat{H} = \frac{1}{\hbar^2} \begin{bmatrix} a & b & 0 & \dots \\ b & a & b & \dots \\ 0 & b & a & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad \text{where} \quad a = \frac{\hbar^2}{m}, \quad b = -\frac{\hbar^2}{2m}$$

When visualising the hydrogen atom, a numerical solver is used for the radial equation, because it contains the Coulomb potential and radial node structure. The spherical harmonics in the angular equation are hard coded using the derived analytical solution. Starting with the radial equation:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} (V(r) - E) = l(l+1)$$

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2 R}{\hbar^2} (V(r) - E) = R l(l+1)$$

$$r^2 R'' + 2r R' - \frac{2mr^2 R}{\hbar^2} (V(r) - E) - R l(l+1) = 0$$

$$R'' + \frac{2}{r} R' - \frac{2m}{\hbar^2} R (V(r) - E) - \frac{R l(l+1)}{r^2} = 0$$

$$-\frac{\hbar^2}{2m} \left(R'' + \frac{2}{r} R' \right) + \left[V(r) + \frac{\hbar^2 l(l+1)}{2mr^2} \right] R = ER$$

This is now in Hamiltonian form ($\hat{H}R = ER$). However, it contains a first derivative term, which would require a further substitution when using the finite difference method. To eliminate it, the substitution $u(r) = rR(r)$ is used.

$$R = \frac{u}{r} \quad R' = \frac{u'}{r} - \frac{u}{r^2} \quad R'' = \frac{u''}{r} - \frac{u'}{r^2} - \frac{u'}{r^2} + \frac{2u}{r^3}$$

$$= \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3}$$

Substituting:

$$-\frac{\hbar^2}{2m} \left(\frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3} + \frac{2u'}{r^2} - \frac{2u}{r^3} \right) + \left[V(r) + \frac{\hbar^2 l(l+1)}{2mr^2} \right] \frac{u}{r} = E \cdot \frac{u}{r}$$

$$-\frac{\hbar^2}{2m} u'' + \left[V(r) + \frac{\hbar^2 l(l+1)}{2mr^2} \right] u = Eu$$

In atomic units, $\hbar = m = e = 1$ and $V(r) = -\frac{1}{r}$. This equation then becomes:

$$-\frac{1}{2} u'' + V_{\text{eff}} u = Eu \quad \text{where} \quad V_{\text{eff}} = -\frac{1}{r} + \frac{l(l+1)}{2r^2}$$

Discretisation involves dividing r into intervals $r_i + ih$ over a range between $r_{\min}$ and $r_{\max}$. Using the finite difference approximation:

$$u_i'' \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}$$

$$u_{i+1} \text{ coefficient: } -\frac{1}{2h^2} \quad u_{i-1} \text{ coefficient: } -\frac{1}{2h^2}$$

$$u_i \text{ coefficient: } \frac{1}{h^2} - \frac{1}{r_i} + \frac{l(l+1)}{2r_i^2}$$